



NSSA 221 Labs
Domain dz6651.com

Systems Administrations I
Site Book

Daniyal Zharov

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Change Control Section

Name	Date	By	Updates
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Creation of Site Book	25/02/2025	Daniyal Zharov	Initial Site Book Creation
Lab 01: RLES Deployment	29/01/2025	Daniyal Zharov	- Launched Solace. - Deployment of the topology. - Configuring PfSense. - Assigning IP addresses to Win22, Win11 and Rocky Linux. - Configured Git Repository on Solace.
Lab 02: Active Directory & Group Policy	25/02/2025	Daniyal Zharov	- Launched Server Manager - Configured the Active Directory, DNS, and DHCP on Win22. - Created DHCP Scope and test user. - Logged into the test user on Win11 and Rocky Linux. - Accessed the Active Directory from test user on Win11. - Created organization units with new users. - Configured Group Policies
Lab 03: RAID and Disk Management	11/03/2025	Daniyal Zharov	- Implemented RAID configurations - formatted disks, managed partitions - tested redundancy solutions - ensured persistent mounts.
Lab 04: File Sharing Services	09/04/2025	Daniyal Zharov	- Installed and configured FTP, Samba, Rsync, and NFS servers on Rocky Linux - Set up user accounts with customized access permissions - Verified Samba and FTP access from Windows and Linux clients - Deployed Rsync module with read-only access and verified transfers using logs - Mounted NFS export on Linux and Windows clients; tested read/write access - Configured firewall and SELinux rules to support all file sharing protocols - Validated persistent mounting using /etc/fstab and Windows registry for UID/GID mapping - Documented client-server connection status using terminal and graphical tools
Lab 05: Cron Jobs and Task Automation	15/04/2025	Daniyal Zharov	- Reviewed crontab structure and created individual user cron entries. - Configured two automated cron tasks using bash scripts: databackup.sh and diskMonitor.sh. - Verified incremental backups and disk usage logs over a period of scheduled execution. - Developed and tested custom bash script networkMonitor.sh for live network availability tracking. - Scheduled network monitor script via crontab and validated output in network_ping.logs.

			- Captured and documented all required screenshots: script listings, cron job validations, and log outputs. - Successfully completed lab objectives and removed cron tasks post-validation.
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Overview

The Site Book captures the setup, deployment, and administrative configuration of a virtual network environment constructed during the NSSA 221. The environment was created within the domain dz6651.com using the Remote Learning Emulation System (RLES). It consists of necessary network functions including Active Directory, DNS, DHCP, RAID storage, file sharing, as well as automation through timed scripts.

The network infrastructure includes a pfSense router, a Windows Server 2022 machine as Domain Controller, DNS server, and DHCP server, a Windows 11 domain client, as well as several Rocky Linux machines for storage, scripts, and for providing services like Samba, FTP, and NFS.

Each of the labs through the course increasingly introduced functionality and complexity:

- Lab 01 involved implementing the virtual environment as well as network address setup.
- Lab 02 presented domain services, Active Directory, and group policies.
- Lab 03 examined RAID configurations as well as mounting of persistent storage with Linux.
- The Lab 04 delved into sharing between platforms through Samba, FTP, NFS, as well as user permissions handling.
- Lab 05 used bash shell scripting and cron jobs for automation of backup, disk monitoring, and network availability.

The Site Book is a living guide for documenting infrastructure choices, troubleshooting configurations, and referencing procedures. It provides an unambiguous technical plan for preserving, rebuilding, or augmenting the network in the long run.

Virtual Environment Setup

The virtualized infrastructure environment was implemented utilizing the Remote Learning Emulation System (RLES) that offered scalable, isolated virtual machines through the RIT network. The deployment environment supported a realistic emulation of actual systems administration by integrating Windows, as well as Linux, operating systems.

Access to the environment was granted via: <https://rlescloud.rit.edu/>

From there, the **vRealize Automation Console** was used to request and manage virtual machines (VMs). RIT credentials were required to authenticate and deploy the infrastructure components.

Table 1. Hostname and Fully Qualified Domain Name (FQDN) Mapping

Device	Hostname	FQDN	Notes
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pfSense Router	gateway	gateway.dz6651.com	Set during pfSense setup (Lab01-Instructions_22451)
Windows Server	winservy	winservy.dz6651.com	Renamed via PowerShell (Lab01-Instructions_22451)
Windows 11 Client	daniyalzh	daniyalzh.dz6651.com	Based on example format (Lab01-Instructions_22451)
Rocky Linux Client	dz6651.student.com	dz6651.student.com	Set via hostnamectl (Lab01-Instructions_22451)
Storage Server	dz6651	dz6651.com	Used in NFS mount, Samba setup (Lab04_Instructions)

Each machine was configured with a static IP address, custom hostname, and integrated into the domain dz6651.com. The **pfSense firewall** provided gateway services, the **Windows Server** managed core domain and IP services, and the **Rocky Linux machines** were used for scripting, file storage, and service configuration.

Git Version Control

The IST department’s “Solace” server maintains a public Git repository, accessible from within the RIT network or via a VPN service. The repository can be accessed using secure shell (SSH) with the following commands:

```
ssh dz6651@solace.ist.rit.edu
```

Essential files like /etc/fstab and /etc/group from storage server as well as configuration and Log files that prove useful for system configuration and maintenance are kept in this Git repository. This becomes a single, version-controlled location to manage and retrieve critical configuration files.

Network Topology and Device Configuration

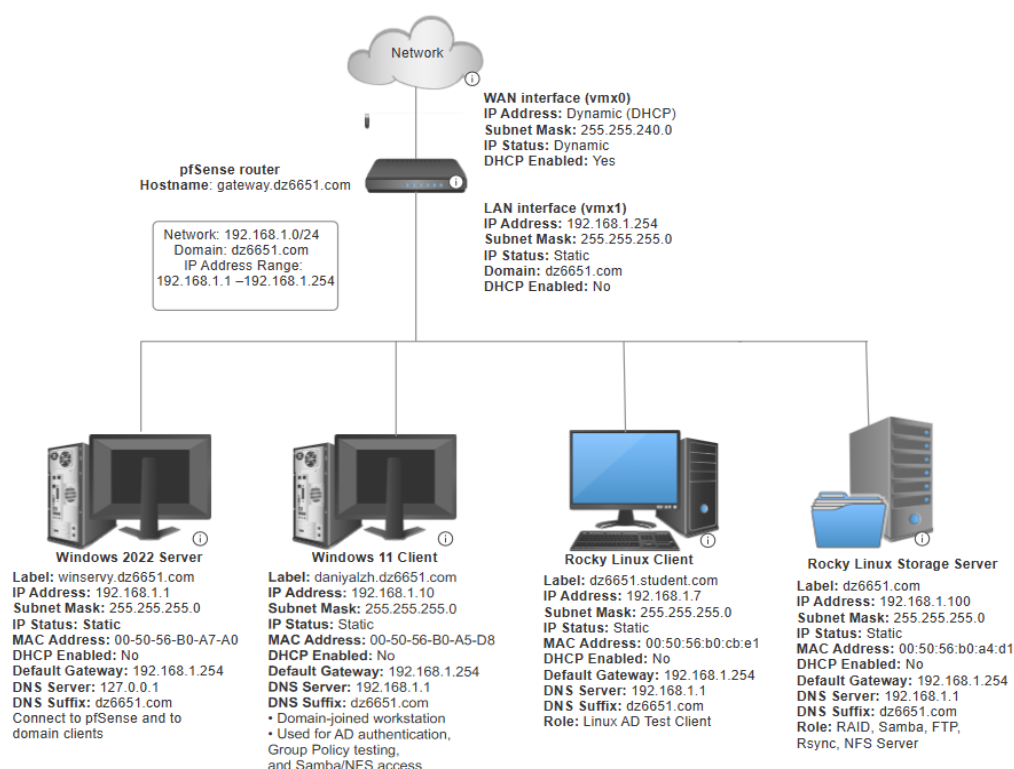
The dz6651.com network topology is composed of a five-machine virtual topology created via the use of RLES. The environment contains a pfSense router, a domain controller running Windows Server, a domain-joined Windows 11 workstation, a Rocky Linux AD test machine, and a Rocky Linux storage server that runs RAID and several file-sharing services. Devices are statically configured within the 192.168.1.0/24 network and are connected via a pfSense-managed virtual LAN.

All of the IPs, hostnames, and roles were configured and checked meticulously using ipconfig /all, hostnamectl, and terminal screenshots directly from Labs 2 to 5. The setup supports domain services centrally, file sharing, redundancy storage, and automated task scheduling across platforms. The following is the table of each device's setup:

Table 2. Devices' Software and Hardware Configuration

Device Name	Type	Hostname	OS	IP Address	Role
pfSense-3966	Router	gateway.dz6651.com	FreeBSD	192.168.1.254	Firewall, NAT, LAN Gateway
WinServer2022-1105	Server	winservy.dz6651.com	Windows Server 2022	192.168.1.1	AD, DNS, DHCP Server
Win11-1171	Workstation	daniyalzh.dz6651.com	Windows 11 Enterprise	192.168.1.10	Domain Workstation
Rocky-Linux-0247	Linux Client	dz6651.student.com	Rocky Linux 9.5	192.168.1.7	Linux AD Test Client
Rocky-Linux-0750	Storage Server	dz6651.com	Rocky Linux 9.5	192.168.1.100	RAID, Samba, NFS, FTP, Rsync

Figure 1. Network topology



Servers

Windows Server 2022

The Windows Server 2022 machine served as the central management server for the dz6651.com domain. It provided Active Directory Domain Services (AD DS), Domain Name System (DNS), and

Dynamic Host Configuration Protocol (DHCP) services to all LAN clients. It was assigned a static IP and connected behind the pfSense router.

Table 3. Windows Server 2022 Settings

Setting	Details
Hostname	winservy
FQDN	winservy.dz6651.com
IP Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254 (pfSense)
Primary DNS	127.0.0.1 (localhost)
Secondary DNS	129.21.3.17 (RIT DNS)
MAC Address	00-50-56-B0-A7-A0
DHCP	No (assigned static IP)
Operating System	Windows Server 2022 Standard

Table 4. Windows Server Roles Installed

Service	Purpose
Active Directory (AD DS)	Centralized identity and access management
DNS Server	Resolves internal domain names (dz6651.com)
DHCP Server	Distributes IP addresses to clients (192.168.1.11–.200)

Rocky Linux Storage Server Configuration

he Rocky Linux Storage Server (dz6651.com) was configured as a high-availability storage and file-sharing platform within the dz6651.com domain. It utilized redundant RAID arrays for disk fault tolerance, and multiple sharing protocols were deployed to ensure cross-platform compatibility between Windows and Linux clients.

This server also performed automated backup and monitoring tasks using custom Bash scripts scheduled via crontab.

Table 5. Storage Server Settings

Setting	Details
Hostname	dz6651
FQDN	dz6651.com

IP Address	192.168.1.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254 (pfSense)
Primary DNS	192.168.1.1 (Windows Server)
Operating System	Rocky Linux 9.5 (Blue Onyx)
MAC Address	00:50:56:B0:A4:D1
DHCP	No (static IP assigned)

Table 6. RAID Configuration

Array	Type	Devices Used	Purpose
RAID 1	Mirroring	/dev/sdc, /dev/sdd	Redundant storage for critical shares
RAID 5	Striping + Parity	/dev/sde, /dev/sdf, /dev/sdg	Balanced performance + redundancy for data storage

Table 7. File Sharing Services Installed

Service	Purpose	Path
Samba (SMB)	Windows file sharing for Ramones/Weezer groups	/media/samba1/ramones
NFS	Linux/Unix file sharing	/media/nfs1/weezer
FTP (vsftpd)	Anonymous upload directory (drop-box)	/var/ftp/uploads/
Rsync Daemon	Automated backup replication	/etc/rsyncd.conf configured

- **Group-based permissions** applied (Ramones group = Samba, Weezer group = NFS)
- **ACLs** configured where necessary

Backup and Monitoring Automation

- **Cron Jobs Scheduled:**
 - networkMonitor.sh — Ping check for network availability
 - diskMonitor.sh — Disk utilization logging
 - databackup.sh — Automated rsync backups to local storage

- Scripts scheduled under crontab for execution every 1–2 minutes
- Output logs stored under /home/student/logs/

RAID and File System Management

The Rocky Linux storage server (dz6651.com) was configured with software-based RAID arrays to provide fault-tolerant storage and support multiple sharing services. RAID arrays were created using the mdadm utility and disks were formatted and persistently mounted using /etc/fstab.

The system uses a logical volume manager (LVM) for the root (/) and swap partitions, while dedicated disks were prepared for RAID configurations.

Figure 2. Storage Server drive configuration

```
[root@dz6651 ~]# hostname; date; which admin
dz6651.com
Mon Mar 10 01:09:36 PM EDT 2025
/usr/bin/which: no admin in (/root/.local/bin:/root/bin:/usr/local/sbin:/sbin:/b
in:/usr/sbin:/usr/bin)
[root@dz6651 ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0     0   40G  0 disk
├─sda1       8:1     0   600M  0 part /boot/efi
├─sda2       8:2     0    1G  0 part /boot
├─sda3       8:3     0  38.4G  0 part
│   └─rl-root 253:0     0  34.5G  0 lvm /
│   └─rl-swap 253:1     0   3.9G  0 lvm [SWAP]
sdb          8:16    0    4G  0 disk /media/samba
sdc          8:32    0    4G  0 disk
sdd          8:48    0    4G  0 disk
sde          8:64    0    4G  0 disk
sdf          8:80    0    4G  0 disk
sdg          8:96    0    4G  0 disk
sdh          8:112   0    4G  0 disk
sdi          8:128   0    4G  0 disk
sr0         11:0    1 1024M  0 rom
[root@dz6651 ~]#
```

Table 8. Storage Devices and Usage

Device	Size	Mount Point	Usage
/dev/sda	40 GB	/ (root), /boot, swap	System Disk
/dev/sdb	4 GB	/media/samba	Samba Share (Standalone)

/dev/sdc, /dev/sdd	4 GB each	RAID 1 Array	Critical Data (Mirrored)
/dev/sde, /dev/sdf, /dev/sdg	4 GB each	RAID 5 Array	Redundant Storage (Striped + Parity)

RAID Configuration

RAID 1 (Mirroring) Array:

- Devices: /dev/sdc, /dev/sdd
- Purpose: Redundant critical data storage
- Created with:

```
mdadm --create /dev/md0 --level=1 --raid-devices=2 /dev/sdc /dev/sdd
```

RAID 5 (Striping with Parity) Array:

- Devices: /dev/sde, /dev/sdf, /dev/sdg
- Purpose: Balanced performance and redundancy
- Created with:

```
mdadm --create /dev/md1 --level=5 --raid-devices=3 /dev/sde /dev/sdf /dev/sdg
```

File System Management

- **File Systems Created:**

RAID Device	File System	Mount Point
/dev/md0	ext4	/media/nfs1 (for NFS)
/dev/md1	xfs	/media/samba1 (for Samba)

Persistent Mounts Using /etc/fstab

RAID devices were configured for automatic mounting by adding UUID references in /etc/fstab.

UUIDs were obtained via: blkid

Figure 3. Persistent Mount Verification

```
[root@dz6651 ~]# hostname; date; mount | grep media; cat /etc/fstab; uptime
dz6651.com
Tue Mar 11 02:31:39 AM EDT 2025
/dev/sdb on /media/samba type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
/dev/md0p1 on /media/nfs1 type ext4 (rw,relatime,seclabel)
/dev/md0p5 on /media/nfs2 type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
/dev/md1p1 on /media/samba1 type ext3 (rw,relatime,seclabel,stripe=256)
/dev/md1p2 on /media/samba2 type ext2 (rw,relatime,seclabel,stripe=256)

#
# /etc/fstab
# Created by anaconda on Mon Apr 10 17:26:26 2023
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rl-root / xfs defaults 0 0
UUID=bbc72667-9448-40ab-a0a9-d83cc98f91c4 /boot xfs defaults 0
0
UUID=6BD1-618E /boot/efi vfat umask=0077,shortname=winnt 0 2
/dev/mapper/rl-swap none swap defaults 0 0
UUID=560b9130-79b9-417b-b738-17c84886ce5d /media/samba xfs defaults 0 0
UUID=0b062e50-dc9b-493a-a28f-69dc086ec3e2 /media/nfs1 ext4 defaults 0 0
UUID=a678c953-c79b-4579-8e90-b9ca7cba7fd5 /media/nfs2 xfs defaults 0 0
UUID=2f944d05-ddca-4549-a57a-9915c1a904ea /media/samba1 ext3 defaults 0 0
UUID=d7ec2bf0-8334-432c-b64f-7fee22d3d187 /media/samba2 ext2 defaults 0 0
02:31:39 up 7:17, 2 users, load average: 0.00, 0.02, 0.00
[root@dz6651 ~]#
```

Validation performed by:

- Running mount -a after edits
- Confirming mounts with df -h
- Reboot testing to confirm automatic re-mount

Ports and Services

Here, the main network ports and services deployed throughout the dz6651.com domain architecture are discussed. Proper handling of ports was imperative for enabling core functionality, secure communications, and ensuring compatibility between the Windows and Linux operating systems.

Firewall settings and services were confirmed using command-line utilities, services status output, and pfSense firewall rules.

Table 9. Core Services and Ports Overview

Service	Port(s)	Protocol	Purpose
DNS (Bind9)	53 (TCP/UDP)	TCP/UDP	Domain Name System resolution

DHCP Server	67 (UDP), 68 (UDP)	UDP	Dynamic IP address assignment
Active Directory (LDAP)	389 (TCP/UDP)	TCP/UDP	AD DS directory services
Active Directory (Kerberos)	88 (TCP/UDP)	TCP/UDP	Authentication ticket granting
SMB/CIFS (Samba)	445 (TCP)	TCP	Windows file sharing
NFS Server	2049 (TCP/UDP)	TCP/UDP	Linux file sharing
FTP (vsftpd)	21 (TCP)	TCP	File Transfer Protocol access
Rsync Daemon	873 (TCP)	TCP	Backup synchronization service
SSH (Optional)	22 (TCP)	TCP	Secure remote access for administration

Windows Server 2022 Services

- **Active Directory Domain Services (AD DS)** enabled ports 389 (LDAP), 88 (Kerberos)
- **DNS Server** handled internal name resolution on port 53
- **DHCP Server** assigned IP addresses on ports 67/68

Rocky Linux Storage Server Services

- **Samba (SMB)** hosted on port 445 for cross-platform file sharing
- **NFS** sharing enabled over port 2049
- **FTP server (vsftpd)** allowed basic file uploads over port 21
- **Rsync Daemon** listening on port 873 for backup jobs
- **SSH** (port 22) available for remote management (secured via internal network only)

Firewall and Security Configuration

- Firewall-cmd and iptables configured on Rocky Linux to allow essential ports only.
- pfSense LAN rules allowed:
 - DNS, DHCP, LDAP, Kerberos from internal network
 - Samba, NFS, FTP, Rsync traffic between storage server and domain clients
- SELinux rules updated where necessary for Samba and NFS access.
- FTP secured for internal use only, not publicly accessible from WAN.

Active Directory

The Active Directory platform was installed on the Windows Server 2022 computer (winsery.dz6651.com) to centralize user authentication, permission management, and allocation of resources across the entire dz6651.com domain.

A new forest was formed while configuring with the root domain being defined as dz6651.com. The server was promoted to the status of Domain Controller employing the interface of the Server Manager. Active Directory Domain Services (AD DS) installation configured services like DNS and integration of DHCP by default

Domain-joined clients included:

- Windows 11 workstation (daniyalzh.dz6651.com)
- Rocky Linux client (dz6651.student.com)

Users and Organizational Units (OUs) Configuration

Test user accounts were created for validating Active Directory functionality across Windows and Linux clients.

Two Organizational Units (OUs) were configured to logically separate users:

- **Ramones**
- **Weezer**

All users were assigned a standard password ("Student@") for academic testing convenience.

Table 10. User Table

Full Name	Logon Name	Password	Organizational Unit (OU)	Domain
Joey Ramone	jramone	Student@	Ramones	dz6651.com
Marky Ramone	marky	Student@	Ramones	dz6651.com
Dee Dee Ramone	deedee	Student@	Ramones	dz6651.com
Johnny Ramone	johnny	Student@	Ramones	dz6651.com
Rivers Cuomo	rcuomo	Student@	Weezer	dz6651.com
Patrick Wilson	pwilson	Student@	Weezer	dz6651.com
Brian Bell	bbell	Student@	Weezer	dz6651.com
Scott Shriner	sshriner	Student@	Weezer	dz6651.com

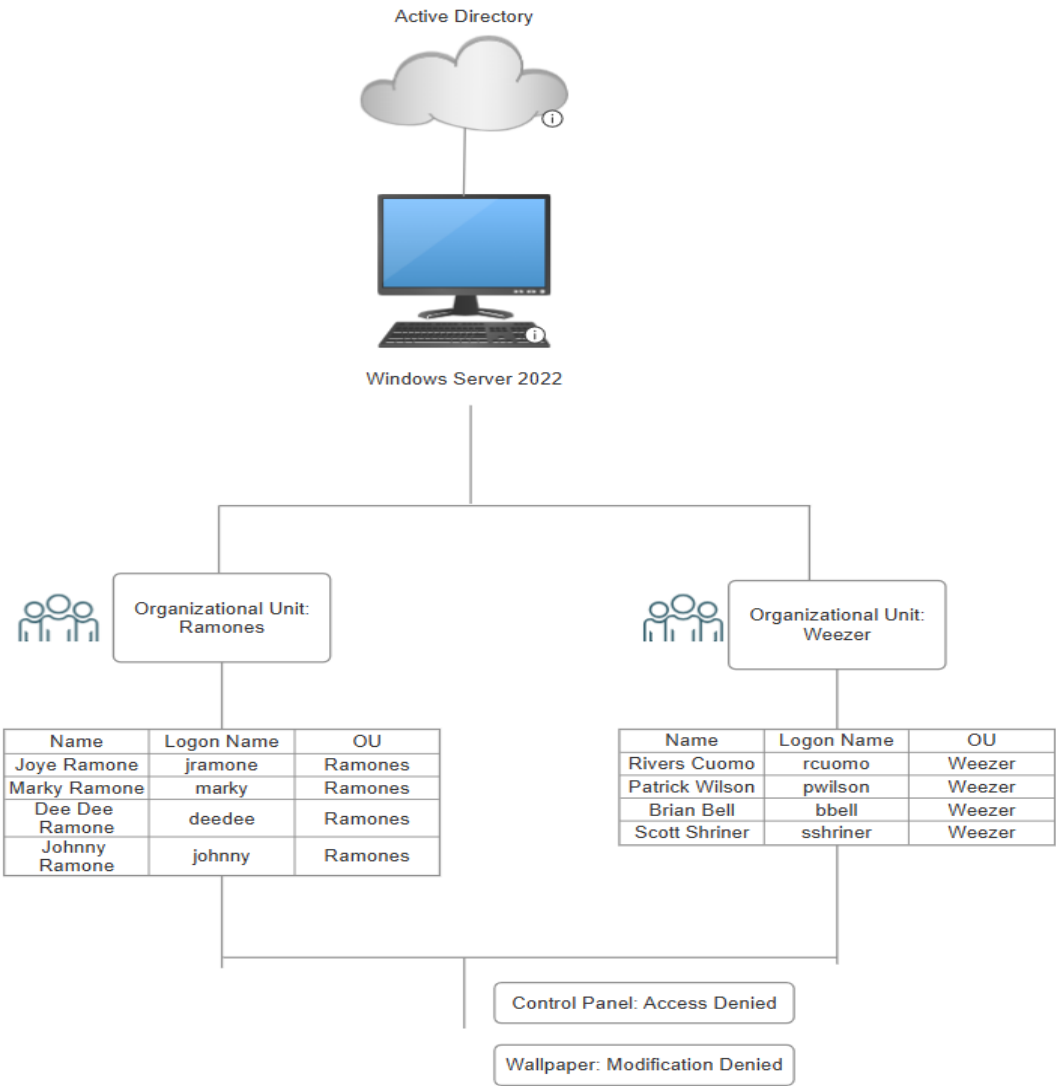
Users and OUs were created using a combination of PowerShell scripting (New-ADUser and New-ADOrganizationalUnit cmdlets) and Windows Admin Center. Automation via PowerShell minimized manual errors and ensured consistency across user creation tasks.

Figure 4. PowerShell History showing Active Directory User and OU Creation

```
[winseriy]: PS C:\Users\tuser\Documents> $env:USERDOMAIN
DZ6651
[winseriy]: PS C:\Users\tuser\Documents> $env:COMPUTERNAME
WINSERY
[winseriy]: PS C:\Users\tuser\Documents> Get-History

Id CommandLine
-----
1 Get-ADDomain
2 New-ADOrganizationalUnit -Name Weezer -Path "dc=gpavks,dc=com" -ProtectedFromAccidentalDeletion:$true -PassThru
3 New-ADOrganizationalUnit -Name Weezer -Path "dc=dz6651,dc=com" -ProtectedFromAccidentalDeletion:$true -PassThru
4 New-ADOrganizationalUnit -Name Weezer -Path "dc=dz6651,dc=com" -ProtectedFromAccidentalDeletion:$true -PassThru
5 New-ADUser -name "Patrick Wilson" -GivenName Patrick - Surname Wilson -UserPrincipalName "pwilson@dz6651.com" -SamAccountName pwilson -Path "ou=weezer,dc=dz6651, dc=com"
6 New-ADUser -name "Patrick Wilson" -GivenName Patrick - Surname Wilson -UserPrincipalName "pwilson@dz6651.com" -SamAccountName pwilson -Path "ou=weezer,dc=dz6651, dc=com"
7 New-ADUser -Name "Patrick Wilson" -GivenName Patrick - Surname Wilson -UserPrincipalName "pwilson@dz6651.com" -SamAccountName pwilson -Path "ou=weezer,dc=dz6651, dc=com"
8 New-ADUser -Name "Rivers Cuomo" -GivenName Rivers -Surname Cuomo -UserPrincipalName "rcuomo@dz6651.com" -SamAccountName rcuomo -Path "ou=weezer,dc=dz6651, dc=com"
9 New-ADUser -Name "Brian Bell" -GivenName Brian -Surname Bell -UserPrincipalName "bbell@dz6651.com" -SamAccountName bbell -Path "ou=weezer,dc=dz6651, dc=com"
10 New-ADUser -Name "Scott Shriner" -GivenName Scott -Surname Shriner -UserPrincipalName "sshriener@dz6651.com" -SamAccountName sshriener -Path "ou=weezer,dc=dz6651, dc=com"
11 Set-ADUser -Identity "CN=Patrick Wilson, OU=weezer, DC=DZ6651,DC=COM" -OfficePhone "5624"
12 Set-ADUser -Identity "CN=Brian Bell, OU=weezer, DC=DZ6651,DC=COM" -OfficePhone "6734"
13 Set-ADUser -Identity "CN=Scott Shriner, OU=weezer, DC=DZ6651,DC=COM" -OfficePhone "5197"
14 Set-ADUser -Identity "CN=River Cuomo, OU=weezer, DC=DZ6651,DC=COM" -OfficePhone "6745"
15 Set-ADUser -Identity "CN=Rivers Cuomo, OU=weezer, DC=DZ6651,DC=COM" -OfficePhone "6745"
16 $env:USERDOMAIN
17 $env:COMPUTERNAME
```

Figure 5. Directory Topology



Group Policy Objects (GPOs)

Two Group Policy Objects (GPOs) were created and applied at the domain level to enforce user restrictions:

GPO Name	Purpose
Control Panel – Access Denied	Disables access to Control Panel and system settings, improving security.
Wallpaper – Modification Denied	Prevents users from changing desktop wallpapers, ensuring branding consistency.

Group Policies were validated by logging into the Windows 11 domain client and observing enforced restrictions.

Dynamic Host Configuration Protocol (DHCP) Setup

The DHCP server was implemented and set up on the Windows Server 2022 computer (winservy.dz6651.com) to automatically provide IP addresses to all domain-based clients within the network dz6651.com.

With dynamic addressing, the DHCP server facilitated easy device management, reduced IP conflicts, and provided seamless network connectivity for Windows as well as Linux devices.

Table 11. DHCP Server Settings

Setting	Details
DHCP Server Host	winservy.dz6651.com
Scope Name	Main Scope
Scope IP Range	192.168.1.11 – 192.168.1.200
Subnet Mask	255.255.255.0
Gateway (Router Option 003)	192.168.1.254
Primary DNS Server (Option 006)	192.168.1.1 (Windows Server)
Secondary DNS Server	129.21.3.17 (RIT DNS)
Lease Duration	8 days

DHCP IP Address Reservation and Exclusions

To ensure critical infrastructure components retained consistent IP addresses and were not assigned dynamically, the following exclusions and static IP assignments were configured:

Device	IP Address	Assignment Type	Notes
pfSense Router	192.168.1.254	Static Manual	Gateway device
Windows Server 2022	192.168.1.1	Static Manual	AD, DNS, DHCP
Rocky Linux Storage Server	192.168.1.100	Static Manual	RAID Storage and File Services
Reserved IP Block	192.168.1.2 – 192.168.1.10	DHCP Exclusion Range	Reserved for critical clients

No dynamic IPs were allowed in .1–.10 or .254 to avoid conflicts.

DHCP Options Configured

- **Option 003 Router:** Set to 192.168.1.254
- **Option 006 DNS Servers:**
 - Primary: 192.168.1.1
 - Secondary: 129.21.3.17

This ensured that all DHCP clients automatically received the correct gateway and DNS configurations upon joining the network.

Testing and Validation

Verified Windows 11 client (daniyalzh.dz6651.com) and Rocky Linux client (dz6651.student.com) both received DHCP leases within the allowed scope.

Used ipconfig /all on Windows and nmcli device show on Linux to confirm address assignments.

Validated correct default gateway and DNS settings received via DHCP.

Domain Name System (DNS) Setup

The Domain Name System (DNS) was set up on the Windows Server 2022 computer (winservy.dz6651.com) while installing Active Directory Domain Services (AD DS).

DNS enables devices under the dz6651.com domain to translate human-language hostnames into their respective IP addresses, making it easy for devices to communicate over the network without having to manually enter IP addresses.

The internal name resolution requests within the LAN are serviced by the Windows Server 2022 running as a DNS server while externally-based queries are redirected from it to a Rochester Institute of Technology (RIT) operated public DNS server.

Table 12. DNS Server Settings

Setting	Details
DNS Server Host	winservy.dz6651.com
Domain	dz6651.com
Primary IP Address	192.168.1.1
Forwarder 1	129.21.3.17 (RIT DNS)
Forwarder 2	(Optional Public DNS e.g., 8.8.8.8 if needed)
Root Hints	Default (Microsoft Root Servers)

DNS Zones Configured

- **Forward Lookup Zone:**
 - Zone Name: dz6651.com
 - Type: Primary Zone (AD-Integrated)
 - Dynamic Updates: Secure Only
 - Devices registered automatically upon domain join (Win11 client, Rocky Linux client)
- **Reverse Lookup Zone:**
 - Zone Name: 1.168.192.in-addr.arpa
 - Type: Primary Zone
 - Purpose: Supports reverse DNS resolution (IP → Hostname lookup)

Table 13. Internal Hostname-to-IP Mappings

Hostname	FQDN	IP Address
pfSense Router	gateway.dz6651.com	192.168.1.254
Windows Server 2022	winservy.dz6651.com	192.168.1.1
Windows 11 Client	daniyalzh.dz6651.com	DHCP-assigned (Dynamic, example 192.168.1.11)
Rocky Linux Client	dz6651.student.com	DHCP-assigned (Dynamic, example 192.168.1.12)
Rocky Linux Storage Server	dz6651.com	192.168.1.100

Dynamic updates were enabled for domain-joined machines, allowing automatic hostname registration after successful domain join and DHCP lease acquisition.

Testing and Validation

- **nslookup** used on Windows and Linux clients to validate internal name resolution:

```
nslookup winservy.dz6651.com
nslookup gateway.dz6651.com
```

- **ping** used to confirm hostname-to-IP communication across LAN:

```
ping storage.dz6651.com
ping dz6651.student.com
```

Dynamic **Updates** observed in DNS Manager after domain join of new clients.

Reverse **Lookup** validated by using IP address queries.

Configuring Local Domain Lookup Using /etc/hosts

In addition to centralized DNS services, manual hostname-to-IP address mappings were configured on each device within the dz6651.com domain using the local /etc/hosts file (Linux) and C:\Windows\System32\drivers\etc\hosts (Windows).

This ensured basic hostname resolution capability even in case of temporary DNS server unavailability, enabling reliable communication between devices during network setup and troubleshooting phases.

Purpose of /etc/hosts Configuration

- Enable immediate hostname resolution without relying on DNS
- Support internal testing (ping, SSH, file sharing) before DNS full deployment
- Provide fallback name resolution mechanism

Table 14. Devices and hostname mapping

Device	Hostname	FQDN	IP Address
pfSense Router	gateway	gateway.dz6651.com	192.168.1.254
Windows Server 2022	winservy	winservy.dz6651.com	192.168.1.1
Windows 11 Client	daniyalzh	daniyalzh.dz6651.com	192.168.1.10
Rocky Linux Client	dz6651.student.com	dz6651.student.com	192.168.1.12
Rocky Linux Storage	dz6651	dz6651.com	192.168.1.100

Editing the Hosts File (Linux and Windows)

On Linux (Rocky Client and Storage Server):

1. Open the host's file:

```
sudo nano /etc/hosts
```

2. Add entries like:

```
192.168.1.1    winservy    winservy.dz6651.com
192.168.1.10  daniyalzh  daniyalzh.dz6651.com
...
```

On Windows (Server and Client):

1. Open Notepad as Administrator.
2. Open:
C:\Windows\System32\drivers\etc\hosts
3. Add entries:

```
192.168.1.1    winservy.dz6651.com
192.168.1.100 dz6651.com
...
```

File Sharing Services Configuration

The Rocky Linux Storage Server (dz6651.com) was configured with multiple file sharing services to enable seamless data exchange between Windows and Linux clients across the dz6651.com domain.

The following services were installed, configured, and tested:

- **Samba (SMB)**
- **NFS (Network File System)**
- **FTP (vsftpd)**
- **Rsync Daemon**

Proper access control, service validation, and firewall adjustments were performed to secure the environment while maintaining functionality.

Samba (SMB) Configuration

Samba was configured to allow Windows clients to access Linux-hosted shares.

- **Service Installed:**

```
sudo dnf install samba samba-client samba-common
```

- **Configuration File Edited:** /etc/samba/smb.conf
- **Sample Share Definitions:**

```
[Ramones]
path = /media/samba1/ramones
writable = yes
browseable = yes
guest ok = no
valid users = @ramones
```

```
[Weezer]
path = /media/nfs1/weezer
writable = yes
browseable = yes
guest ok = no
valid users = @weezer
```

- **User and Group Setup:**

```
sudo groupadd ramones
sudo groupadd weezer
sudo usermod -aG ramones jramone
sudo usermod -aG weezer rcuomo
sudo smbpasswd -a [username]
```

- **Firewall Adjustments:**

```
sudo firewall-cmd --add-service=samba --permanent
sudo firewall-cmd --reload
```

- **Testing:**

- Windows client accessed Samba share via `\\dz6651.com\Ramones`
- Linux client accessed via `smbclient`

NFS (Network File System) Configuration

NFS was configured to allow Linux clients to mount shares from the storage server.

- **Service Installed:**

```
sudo dnf install nfs-utils
```

- **Configuration:** `/etc/exports`

```
/media/nfs1/weezer *(rw,sync,no_root_squash)
```

- **Starting Services:**

```
sudo systemctl enable nfs-server
sudo systemctl start nfs-server
```

- **Firewall Adjustments:**

```
sudo firewall-cmd --add-service=nfs --permanent
sudo firewall-cmd --reload
```

- **Testing:**

Linux clients mounted NFS share using:

```
sudo mount dz6651.com:/media/nfs1/weezer /mnt/testmount
```

FTP Server (vsftpd) Configuration

A basic FTP server was configured for uploading files anonymously to the storage server.

- **Service Installed:**

```
sudo dnf install vsftpd
```

- **Anonymous Upload Directory Created:** /var/ftp/uploads
- **vsftpd Configuration File Edited:** /etc/vsftpd/vsftpd.conf

```
anonymous_enable=YES
write_enable=YES
anon_upload_enable=YES
anon_mkdir_write_enable=YES
```

- **SELinux Context Adjustments:**

```
sudo setsebool -P ftp_home_dir=1
```

- **Firewall Adjustments:**

```
sudo firewall-cmd --add-service=ftp --permanent
sudo firewall-cmd --reload
```

- **Testing:**

Windows and Linux clients uploaded files anonymously to the FTP server.

Rsync Daemon Configuration

Rsync was configured to allow incremental backups and synchronization across systems.

- **Service Installed:**

```
sudo dnf install rsync
```

- **Configuration File:** /etc/rsyncd.conf

```
pid file = /var/run/rsyncd.pid
lock file = /var/run/rsync.lock
log file = /var/log/rsyncd.log
```

```
[backup]
path = /media/sambal/ramones
comment = Backup Share
read only = no
list = yes
uid = root
gid = root
auth users = backupuser
secrets file = /etc/rsyncd.secrets
```

- **Starting Service:**

```
sudo systemctl enable rsyncd
sudo systemctl start rsyncd
```

- **Firewall Adjustments:**

```
sudo firewall-cmd --add-port=873/tcp --permanent
sudo firewall-cmd --reload
```

Crontab and Backup Automation

To automate network monitoring, backup tasks, and disk utilization logging, custom Bash scripts were developed and scheduled via user-level crontab entries on the Rocky Linux Storage Server (dz6651.com).

This setup ensured consistent backup operations, proactive network status tracking, and continuous disk usage monitoring without manual intervention.

Table 15. Scripts Developed

Script Name	Purpose
databackup.sh	Automate incremental backups of Documents to the RAID storage share
networkMonitor.sh	Monitor network availability by pinging pfSense gateway
diskMonitor.sh	Record disk space usage statistics periodically

Script Locations

All scripts were placed under:

```
/home/student/scripts/
```

Scripts were made executable using:

```
chmod +x /home/student/scripts/*.sh
```

Crontab Scheduling

Each script was scheduled using the crontab -e editor under the **student** user.

Example crontab entries:

```
# Backup home directory every 1 minute
*/1 * * * * /home/student/scripts/databackup.sh

# Monitor network connection every 2 minutes
*/2 * * * * /home/student/scripts/networkMonitor.sh#
```

Ping check every 1 minutes

```
*/1 * * * * /home/student/scripts/pingcheck.sh
```

Details of Each Script

databackup.sh

```
rsync -avh /home/student/Documents/ /media/samba1/backup/ >>  
/home/student/logs/databackup.log 2>&1
```

- Backs up Documents folder to /media/samba1/backup
- Appends rsync operation results to /home/student/logs/databackup.log

networkMonitor.sh

```
ping -c 3 192.168.1.254 >> /home/student/logs/network_ping.logs 2>&1
```

- Pings the pfSense gateway
- Logs results (success or packet loss) to /home/student/logs/network_ping.logs

diskMonitor.sh

```
df -h >> /home/student/logs/disk_usage.logs 2>&1
```

- Captures current disk usage snapshot
- Appends results to /home/student/logs/disk_usage.logs

Testing and Validation

- Scripts were **manually executed** to ensure correct functionality.
- `crontab -l` was used to verify scheduled jobs were active.
- **Log files** were checked to confirm periodic execution:
 - /home/student/logs/databackup.log
 - /home/student/logs/network_ping.logs
 - /home/student/logs/disk_usage.logs
- NetworkMonitor script was validated by simulating network disruptions

Backup Data Verification

- The backup destination /media/samba1/backup/ was inspected.
- New copies of the /home/student/Documents/ folder were found under the backup directory.

- Incremental file copies confirmed rsync operated correctly without overwriting unchanged files.

Network Failure Testing

- Simulated network failure by disabling pfSense WAN interface temporarily.
- `networkMonitor.sh` correctly logged failed ping attempts in `/home/student/logs/network_ping.logs`.
- Upon network restoration, successful pings were logged, demonstrating operational monitoring.

Disk Utilization Monitoring Validation

- Checked `/home/student/logs/disk_usage.logs`.
- Confirmed that disk space snapshots were appended consistently every 2 minutes.
- Growth of the log file over time validated that the script was operating automatically through `crontab`.

Appendix: Script Listings

Script 1: Ping Check Script (pingcheck.sh)

```
#!/bin/bash
```

```
TARGET_IP="192.168.5.99"
```

```
LOG_FILE="/home/student/network_ping.logs"
```

```
DATE_TIME=$(date)
```

```
HOST=$(hostname)
```

```
ping -c 1 $TARGET_IP > /dev/null 2>&1
```

```
if [ $? -eq 0 ]; then
```

```
    echo "$TARGET_IP Connection Alive $DATE_TIME $HOST" >> "$LOG_FILE"
```

```
else
```

```
    echo "$TARGET_IP Connection Failure $DATE_TIME $HOST" >>  
    "$LOG_FILE"
```

```
fi
```

Script 2: Data Backup Script (databackup.sh)

```
#!/bin/bash

BACKUP_DIR="/home/student/databackup/"

mkdir -p "$BACKUP_DIR"

COUNTER_FILE="$BACKUP_DIR/backup_counter.txt"
if [ ! -f "$COUNTER_FILE" ]; then
    echo "1" > "$COUNTER_FILE"
fi

COUNTER=$(cat "$COUNTER_FILE")
NEW_BACKUP_DIR="$BACKUP_DIR/data_backup_$COUNTER"
mkdir -p "$NEW_BACKUP_DIR"

rsync -a --delete /home/student/data/ "$NEW_BACKUP_DIR/"

COUNTER=$((COUNTER + 1))
echo "$COUNTER" > "$COUNTER_FILE"
```

Script 3: Disk Utilization Monitor Script (diskMonitor.sh)

```
#!/bin/bash

LOG_FILE="/home/student/logs/disk_usage.log"

echo "Date: $(date)" >> "$LOG_FILE"
echo "Hostname: $(hostname)" >> "$LOG_FILE"
df -h >> "$LOG_FILE"
echo " " >> "$LOG_FILE"
```